A WKB-based fixed-grid method for capturing trait concentration in a dispersal evolution model

Weijie Huang* and Xinran Ruan†

Abstract

The evolution of dispersal traits is a fundamental topic in evolutionary ecology, where natural selection may drive the trait distribution toward concentration in the rare mutation regime. This singular behavior poses a serious numerical difficulty, since direct discretizations of the population density require very fine trait grids to identify the fittest trait and to resolve the concentrated profile accurately. In this paper, we develop a WKB-based numerical framework for a dispersal evolution model. By separating the exponentially concentrated trait dependence from a smoother amplitude variable and combining this WKB representation with specially designed reconstruction techniques, the method recovers the selected trait and the associated concentration structure accurately and efficiently on fixed trait grids. We establish a semi-discrete stationary fixed-grid asymptotic-preserving structure for the rare-mutation limit of the steady-state problem. Numerical experiments compare the proposed method with direct density discretizations and confirm its advantage in the small mutation regime.

Key words. dispersal evolution, rare mutation limit, trait concentration, WKB reformulation, Hamilton–Jacobi limit, asymptotic-preserving structure.

MSC 2020. 65M06, 65M12, 35B25, 35F21, 92D15.

1 Introduction

Adaptive dynamics studies the evolution of phenotypic traits under mutation, selection, and ecological feedback. In structured population models, natural selection is often manifested by the concentration of the population density on one or several fittest traits. In the rare-mutation regime, such concentration is usually described by a measure-valued limit in the trait variable, and the support of the limiting measure represents the selected phenotype. This viewpoint has been developed in selection-mutation models and in the Hamilton–Jacobi approach to adaptive dynamics [3, 6]. Among these problems, the evolution of dispersal is a particularly important example. In spatially heterogeneous environments, different dispersal rates may lead to different fitness advantages, and the selected strategy depends on the interaction between movement, resource distribution, and population competition. This question has been studied from several viewpoints in mathematical ecology, for example in reaction-diffusion models for slow dispersal, ideal free distributions, and spatial sorting [7, 9, 19].

In this work, we consider a representative dispersal evolution model of the form

$$\varepsilon\partial_t n_\varepsilon - D(\theta)\Delta_{\mathbf{x}} n_\varepsilon - \varepsilon^2\Delta_\theta n_\varepsilon = n_\varepsilon\big(K(\mathbf{x}) - \rho_\varepsilon(\mathbf{x})\big), \qquad \mathbf{x}\in\Omega. \tag{1.1}$$

*School of Mathematics and Statistics, Beijing Jiaotong University, Beijing 100044, People's Republic of China (wjhuang@bjtu.edu.cn)

†School of Mathematical Sciences, Capital Normal University, Beijing, China, 100048, People's Republic of China (xinran.ruan@cnu.edu.cn).

Here t denotes time, $\mathbf{x}$ is the spatial variable, and θ is the phenotypic trait. The unknown $n_\varepsilon(t, \mathbf{x}, \theta)$ is the structured population density, and

$$\rho_\varepsilon(t, \mathbf{x}) = \int_{\mathbb{T}} n_\varepsilon(t, \mathbf{x}, \theta)\, d\theta, \quad P_\varepsilon(t, \theta) = \int_\Omega n_\varepsilon(t, \mathbf{x}, \theta)\, d\mathbf{x},$$

denote the population density in space and the trait marginal, respectively. The concentration of P_ε in the rare-mutation regime is used to identify the selected trait. The function $K(\mathbf{x})$ represents the local carrying capacity, while the trait-dependent diffusivity $D(\theta)$ models the dispersal rate associated with phenotype θ. Throughout the paper, we assume that the dispersal rate is positive and periodic in the trait variable and that the carrying capacity is nonnegative and bounded:

$$0 < D_{\min} \le D(\theta) \le D_{\max} < \infty, \qquad 0 \le K(\mathbf{x}) \le K_{\max}. \tag{1.2}$$

We impose homogeneous Neumann boundary conditions on $\partial\Omega$ and periodic boundary conditions in the trait direction. The time-dependent model (1.1) is used here as an evolutionary relaxation toward the selection state, while the asymptotic structure of interest is motivated by the rare-mutation limit of the associated steady-state problem.

The steady-state version of this dispersal evolution model was studied by B. Perthame and P. E. Souganidis in [18]. Their work provides a rare-mutation description of the selected dispersal trait for a population structured by space and by a continuous dispersal trait. In the limit of vanishing mutations, the population concentrates in the trait variable, while the spatial profile remains nontrivial. This result gives the asymptotic background for the present paper and clarifies the limiting selection structure that the numerical method aims to capture.

This dispersal model fits into a broader class of structured selection models with concentration in the small-mutation limit. For purely trait-structured or nonlocal selection models, Dirac concentration and constrained Hamilton–Jacobi equations have been studied in [5] and [15]. When space is included as an additional structuring variable, the analysis becomes more delicate. Related Hamilton–Jacobi limits for populations structured by space and trait were obtained in [4], and an asymptotic analysis of a selection model with space was given in [17]. The Perthame–Souganidis dispersal model has also motivated later studies on dispersal evolution, conditional dispersal, and the stability of Dirac concentrations [8, 13, 14].

The same concentration mechanism creates a severe numerical difficulty. Direct discretizations of the original density must resolve a trait profile whose width shrinks with the mutation scale. Hence the mesh required in the trait direction may become prohibitively fine when ε is small. This suggests that the numerical method should exploit the asymptotic separation between the singular trait dependence and the smoother amplitude, rather than resolve the concentrated density directly.

This is closely related to the idea of asymptotic-preserving schemes, whose goal is to capture the limiting regime on discretizations that do not resolve the small scale ε [11]. In rare-mutation problems, the natural asymptotic separation is provided by a WKB or Hopf–Cole transformation. The density is represented by an exponential phase, which describes the concentration in the trait variable, multiplied by a smoother amplitude. Such a WKB viewpoint is standard in the Hamilton–Jacobi approach to adaptive dynamics [3] and is also central in the dispersal model of [18].

WKB-type transformations have been used as a numerical tool in several multiscale problems with singular or highly oscillatory structures. In adaptive dynamics, a WKB representation was used in [1] to construct an asymptotic-preserving scheme that captures Dirac concentrations on coarse, ε-independent phenotype meshes. Related Hopf–Cole based AP ideas have also been used for kinetic reaction-transport equations in singular front propagation regimes [10]. In kinetic BGK models, a Hopf–Cole transformation has been used to approximate the density through an effective phase equation and higher order correction terms [16]. WKB-based numerical ideas are also well established for highly oscillatory semiclassical problems, where a WKB-type preprocessing separates the dominant oscillations and allows the remaining smoother problem to be discretized on much coarser grids [2, 12].

The present paper develops a WKB-based fixed-grid numerical framework for the dispersal evolution problem (1.1). Starting from the phase-amplitude representation, we derive a semi-discrete scheme for the phase and amplitude variables and prove the positivity of the amplitude. We then establish a stationary fixed-grid AP structure associated with the rare-mutation limit of the steady-state problem under a one-sided remainder stability condition. At the fully discrete level, the time-marching scheme is coupled with a scalar mass-balance projection and a reconstruction layer. The reconstruction layer recovers the spatial marginal, the trait marginal, and the local concentration structure from the WKB variables. It is essential for accurate small-ε computations and also supports the dual trait-grid implementation, in which the one-dimensional phase is resolved on a finer trait grid while the space-dependent amplitude is evolved on a coarser grid.

The paper is organized as follows. Section 2 recalls the WKB ansatz and the steady-state rare-mutation structure that motivates the numerical formulation. Section 3 presents the semi-discrete WKB scheme, proves positivity of the amplitude, and establishes the stationary fixed-grid AP structure. Section 4 is devoted to the fully discrete method, including the scalar mass-balance projection, the reconstruction of density-level quantities, and the dual trait-grid implementation. Section 5 contains numerical experiments comparing the WKB method with direct density discretizations and testing the reconstruction of the selected trait in the small-mutation regime. The appendices collect the auxiliary numerical ingredients and implementation details used in the main text.

2 Preliminary

2.1 WKB ansatz and reformulated system

To capture the concentration phenomenon in the trait variable, we introduce the classical WKB ansatz

$$n_\varepsilon(x,\theta,t) = W_\varepsilon(x,\theta,t)\,\exp\left(\frac{u_\varepsilon(\theta,t)}{\varepsilon}\right). \tag{2.3}$$

This decomposition separates the exponentially varying part in the trait direction from a comparatively smoother amplitude. It is therefore natural for both the continuous asymptotic analysis and the numerical design in the small-mutation regime.

Substituting (2.3) into (1.1) and assigning the leading trait-direction exponential contributions to the phase function, we obtain the reformulated system

$$\begin{cases} \partial_t u_\varepsilon - |\partial_\theta u_\varepsilon|^2 - \varepsilon\,\partial_{\theta\theta} u_\varepsilon = -\mathcal{H}(\theta,t), \\ \varepsilon\partial_t W_\varepsilon - D(\theta)\Delta_x W_\varepsilon - \varepsilon^2\partial_{\theta\theta} W_\varepsilon - 2\varepsilon\,\partial_\theta W_\varepsilon\,\partial_\theta u_\varepsilon = W_\varepsilon\big(K(x) - \rho_\varepsilon(x,t) + \mathcal{H}(\theta,t)\big), \end{cases} \tag{2.4}$$

where $\mathcal{H}(\theta,t)$ denotes an effective Hamiltonian. Using the identity

$$\partial_\theta(W_\varepsilon\partial_\theta u_\varepsilon) = \partial_\theta W_\varepsilon\,\partial_\theta u_\varepsilon + W_\varepsilon\partial_{\theta\theta} u_\varepsilon,$$

we rewrite the transport coupling in conservative form. If

$$F_\varepsilon = -W_\varepsilon\partial_\theta u_\varepsilon, \tag{2.5}$$

then

$$-2\varepsilon\partial_\theta W_\varepsilon\,\partial_\theta u_\varepsilon = 2\varepsilon\partial_\theta F_\varepsilon + 2\varepsilon W_\varepsilon \partial_{\theta\theta} u_\varepsilon.$$

Consequently, the system can be written as

$$\begin{cases} \partial_t u_\varepsilon - |\partial_\theta u_\varepsilon|^2 - \varepsilon\,\partial_{\theta\theta} u_\varepsilon = -\mathcal{H}(\theta,t), \\ \varepsilon\partial_t W_\varepsilon - D(\theta)\Delta_x W_\varepsilon - \varepsilon^2\partial_{\theta\theta} W_\varepsilon + 2\varepsilon\partial_\theta F_\varepsilon = W_\varepsilon\big(K(x) - \rho_\varepsilon(x,t) + \mathcal{H}(\theta,t) - 2\varepsilon\partial_{\theta\theta} u_\varepsilon\big). \end{cases} \tag{2.6}$$

In the continuous theory, $\mathcal{H}$ is related to a principal eigenvalue problem in the spatial variable. This structure is the basic reason for introducing the variables $(u_\varepsilon, W_\varepsilon)$ in the first place.

2.2 Continuous asymptotic structure of the steady state

We next recall the steady-state asymptotic structure that motivates the numerical method. Let $(\bar n_\varepsilon, \bar\rho_\varepsilon)$ be a positive steady state of (1.1). Then

$$-D(\theta)\Delta_x \bar n_\varepsilon - \varepsilon^2\partial_{\theta\theta}\bar n_\varepsilon = \bar n_\varepsilon\big(K(x) - \bar\rho_\varepsilon(x)\big), \qquad \bar\rho_\varepsilon(x) = \int_{\mathbb{T}} \bar n_\varepsilon(x,\theta)\,d\theta. \tag{2.7}$$

As shown in [18], under standard assumptions on K, D, and the domain, the steady-state family admits several properties that are fundamental for the rare-mutation limit.

Proposition 2.1 (Continuous asymptotic structure of the steady state)**.** *Assume that K is positive and nonconstant, that D is positive, periodic, and has a unique minimizer θ_m, and that the steady problem* (2.7) *admits a positive solution. Then the following properties hold at the continuous level.*

1. *There exists a constant $C > 0$, independent of ε, such that*

$$0 \le \bar\rho_\varepsilon(x) \le C \qquad \textit{for all } x \in \Omega. \tag{2.8}$$

   Moreover, up to subsequences, $\bar\rho_\varepsilon$ converges uniformly to a limit density ρ.

2. *If we write*

$$\bar n_\varepsilon(x,\theta) = \exp\left(\frac{\bar u_\varepsilon(x,\theta)}{\varepsilon}\right), \tag{2.9}$$

   then $\bar u_\varepsilon$ is uniformly Lipschitz continuous in θ, and along subsequences

$$\bar u_\varepsilon \to u \qquad \textit{uniformly in } (x,\theta), \tag{2.10}$$

   where the limit u is independent of x, periodic in θ, and satisfies

$$\begin{cases} |\partial_\theta u(\theta)|^2 = \mathcal{H}\big(\theta, \rho(\cdot)\big), \\ \max\limits_{\theta\in\mathbb{T}} u(\theta) = 0. \end{cases} \tag{2.11}$$

3. *The effective Hamiltonian $\mathcal{H}(\theta,\rho)$ is defined through the principal eigenvalue problem*

$$\begin{cases} -D(\theta)\Delta_x \mathcal{N}(x,\theta) = \mathcal{N}(x,\theta)\big(K(x) - \rho(x)\big) + \mathcal{N}(x,\theta)\,\mathcal{H}\big(\theta,\rho(\cdot)\big), & x \in \Omega, \\ \partial_\nu \mathcal{N} = 0, & x \in \partial\Omega, \end{cases} \tag{2.12}$$

   with a positive eigenfunction $\mathcal{N}$. Moreover, $\mathcal{H}$ has the same monotonicity in θ as D, and hence

$$\min_{\theta\in\mathbb{T}} \mathcal{H}\big(\theta,\rho(\cdot)\big) = \mathcal{H}\big(\theta_m, \rho(\cdot)\big) = 0. \tag{2.13}$$

4. The steady state concentrates on the fittest trait in the limit $\varepsilon \to 0$. More precisely,

$$\bar{n}_\varepsilon \rightharpoonup N_m(x)\,\delta(\theta - \theta_m), \qquad \bar{\rho}_\varepsilon \to N_m(x), \tag{2.14}$$

where N_m solves the reduced Fisher-type problem

$$\begin{cases} -D(\theta_m)\Delta_x N_m = N_m\big(K(x) - N_m\big), & x \in \Omega, \\ \partial_\nu N_m = 0, & x \in \partial\Omega. \end{cases} \tag{2.15}$$

Proposition 2.1 provides the continuous target of the numerical scheme. The boundedness of $\bar{\rho}_\varepsilon$ keeps the effective Hamiltonian in a bounded regime. The constrained Hamilton–Jacobi equation identifies the dominant trait through the maximizer of u. The concentration result (2.14) indicates that an asymptotic-preserving discretization should recover the reduced pair (N_m, θ_m) without resolving the original density on an extremely fine trait grid. This is the sense in which the WKB reformulation is used in the sequel.

3 WKB reformulated numerical scheme

In this section, we develop and analyze a semi-discrete numerical scheme for the WKB reformulation (2.6). The scheme is formulated in terms of the phase variable u_ε and the amplitude variable W_ε, with the total density n_ε reconstructed from the WKB representation.

3.1 Semi-discrete numerical discretization

We work at the semi-discrete level in order to separate the discretization in (x, θ) from the additional issue of time stepping. For clarity, we present the scheme in one spatial dimension with uniform grids in the x- and θ-directions. This choice is not essential for the WKB decomposition, but it keeps the notation transparent.

Let $\{x_j\}_{j=1}^{J}$ be a uniform grid on $\Omega = (a, b)$ and let $\{\theta_k\}_{k=1}^{K}$ be a uniform periodic grid on $\mathbb{T} = (0, 1)$, with mesh size $\Delta\theta$. We denote

$$W_{j,k}^{\varepsilon}(t) \approx W_\varepsilon(t, x_j, \theta_k), \qquad u_k^{\varepsilon}(t) \approx u_\varepsilon(t, \theta_k).$$

For notational convenience, we write

$$D_k = D(\theta_k), \qquad K_j = K(x_j).$$

By (1.2), we have

$$0 < D_{\min} \le D_k \le D_{\max} < \infty, \qquad 0 \le K_j \le K_{\max}.$$

For a grid function $V = \{V_{j,k}\}$, we use the standard second-order difference operators

$$\delta_x^2 V_{j,k} = \frac{V_{j+1,k} - 2V_{j,k} + V_{j-1,k}}{(\Delta x)^2}, \qquad \delta_\theta^2 V_{j,k} = \frac{V_{j,k+1} - 2V_{j,k} + V_{j,k-1}}{(\Delta\theta)^2}.$$

The homogeneous Neumann boundary condition in x is imposed by the ghost values

$$V_{0,k} = V_{1,k}, \qquad V_{J+1,k} = V_{J,k},$$

while all indices in the θ-direction are understood periodically. For the phase variable, we also use

$$D_\theta^- u_k = \frac{u_k - u_{k-1}}{\Delta\theta}, \qquad D_\theta^+ u_k = \frac{u_{k+1} - u_k}{\Delta\theta}.$$

The semi-discrete WKB system involves two numerical ingredients that are not contained in the standard second-order difference operators introduced above. The first one is the numerical Hamiltonian for the phase equation. The Hamilton–Jacobi term $-|\partial_\theta u_\varepsilon|^2$ is approximated by

$$\widehat{H}(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon).$$

We assume that $\widehat{H}$ is locally Lipschitz and consistent with the continuous Hamiltonian, namely

$$\widehat{H}(p,p) = -p^2.$$

We also assume that it is monotone in the sense that it is nondecreasing with respect to its first argument and nonincreasing with respect to its second argument. When $\widehat{H}$ is differentiable, this condition reads

$$\partial_a \widehat{H}(a,b) \geq 0, \qquad \partial_b \widehat{H}(a,b) \leq 0.$$

Typical examples are the Godunov and Lax–Friedrichs numerical Hamiltonians, as recalled in Appendix **??**.

The second ingredient is the conservative discretization of the transport coupling in the amplitude equation. Recall that the continuous flux is

$$F_\varepsilon = -W_\varepsilon \partial_\theta u_\varepsilon.$$

We denote by

$$\mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k}$$

a conservative discretization of $\partial_\theta F_\varepsilon$ at (x_j, θ_k). Here $\widehat{\mathcal{F}}$ approximates the flux $-W_\varepsilon \partial_\theta u_\varepsilon$. We assume that $\mathcal{D}_\theta \widehat{\mathcal{F}}$ is consistent with

$$\partial_\theta F_\varepsilon = \partial_\theta(-W_\varepsilon \partial_\theta u_\varepsilon)$$

when evaluated on smooth functions, and that it is conservative in the periodic trait variable. For the positivity argument below, we further assume the quasi-positivity property

$$W_{j,k} = 0, \qquad W_{j,\ell} \geq 0 \text{ for all } \ell \quad \Longrightarrow \quad -\mathcal{D}_\theta \widehat{\mathcal{F}}(W,u)_{j,k} \geq 0. \tag{3.16}$$

This property is satisfied by standard monotone finite volume fluxes, such as the upwind and Lax–Friedrichs fluxes listed in Appendix **??**.

With these two ingredients specified, we now define the discrete density and the semi-discrete WKB system. For notational convenience, set

$$E_k^\varepsilon(t) = \exp\left(\frac{u_k^\varepsilon(t)}{\varepsilon}\right). \tag{3.17}$$

The reconstructed nodal density is defined by

$$n_{j,k}^\varepsilon(t) = W_{j,k}^\varepsilon(t) E_k^\varepsilon(t). \tag{3.18}$$

As we shall see later, the evolution of u_ε and W_ε is coupled through the spatial marginal density. In the semi-discrete scheme, this marginal is computed by the composite quadrature in the trait direction. More precisely, we define

$$\rho_{j,Q}^\varepsilon(t) = m_j^\varepsilon(t) := \Delta\theta \sum_{k=1}^{K} W_{j,k}^\varepsilon(t) E_k^\varepsilon(t), \qquad j = 1, \ldots, J. \tag{3.19}$$

This is the discrete counterpart of

$$\int_{\mathbb{T}} n_\varepsilon(t, x_j, \theta)\, d\theta = \int_{\mathbb{T}} W_\varepsilon(t, x_j, \theta) \exp\left(\frac{u_\varepsilon(t,\theta)}{\varepsilon}\right)\, d\theta.$$

Given

$$\boldsymbol{\rho}_Q(t) = \big(\rho^\varepsilon_{1,Q}(t), \dots, \rho^\varepsilon_{J,Q}(t)\big),$$

the discrete effective Hamiltonian $\mathcal{H}_k(\boldsymbol{\rho}_Q(t))$ is determined by the discrete principal eigenvalue problem

$$-D_k \delta_x^2 \mathcal{N}_j^{(k)} = \mathcal{N}_j^{(k)} \big(K_j - \rho^\varepsilon_{j,Q}(t)\big) + \mathcal{N}_j^{(k)} \mathcal{H}_k\big(\boldsymbol{\rho}_Q(t)\big), \qquad j = 1, \dots, J. \tag{3.20}$$

Here $\mathcal{N}^{(k)}$ is the corresponding positive eigenvector, equipped with the same discrete Neumann boundary condition in x. Its normalization does not affect $\mathcal{H}_k$. For definiteness, one may impose

$$\Delta x \sum_{j=1}^{J} \mathcal{N}_j^{(k)} = 1.$$

With the above notation, the semi-discrete WKB scheme reads

$$\begin{cases} \dfrac{\mathrm{d}}{\mathrm{d}t} u_k^\varepsilon + \widehat{H}\left(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon\right) - \varepsilon \delta_\theta^2 u_k^\varepsilon = -\mathcal{H}_k\big(\boldsymbol{\rho}_Q(t)\big), \qquad k = 1, \dots, K, \\ \varepsilon \dfrac{\mathrm{d}}{\mathrm{d}t} W_{j,k}^\varepsilon - D_k \delta_x^2 W_{j,k}^\varepsilon - \varepsilon^2 \delta_\theta^2 W_{j,k}^\varepsilon + 2\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k} \\ \qquad = W_{j,k}^\varepsilon \big(K_j - \rho^\varepsilon_{j,Q}(t) + \mathcal{H}_k(\boldsymbol{\rho}_Q(t)) - 2\varepsilon \delta_\theta^2 u_k^\varepsilon\big), \qquad j = 1, \dots, J, \quad k = 1, \dots, K. \end{cases} \tag{3.21}$$

We next show the equation satisfied by the reconstructed density $n_{j,k}^\varepsilon$. Since E_k^ε is independent of the spatial index j,

$$\delta_x^2 n_{j,k}^\varepsilon = E_k^\varepsilon \delta_x^2 W_{j,k}^\varepsilon.$$

Moreover,

$$\varepsilon \frac{\mathrm{d}}{\mathrm{d}t} n_{j,k}^\varepsilon = E_k^\varepsilon \left(\varepsilon \frac{\mathrm{d}}{\mathrm{d}t} W_{j,k}^\varepsilon + W_{j,k}^\varepsilon \frac{\mathrm{d}}{\mathrm{d}t} u_k^\varepsilon\right). \tag{3.22}$$

Multiplying the first equation in (3.21) by $E_k^\varepsilon W_{j,k}^\varepsilon$ and multiplying the second equation in (3.21) by $\varepsilon E_k^\varepsilon$, noticing (3.22), we can derive that $n_{j,k}^\varepsilon$ satisfies

$$\varepsilon \frac{\mathrm{d}}{\mathrm{d}t} n_{j,k}^\varepsilon - D_k \delta_x^2 n_{j,k}^\varepsilon = \big(K_j - \rho^\varepsilon_{j,Q}\big) n_{j,k}^\varepsilon + R_{j,k}^\varepsilon, \tag{3.23}$$

where the remainder has the form

$$R_{j,k}^\varepsilon = \varepsilon^2 E_k^\varepsilon \delta_\theta^2 W_{j,k}^\varepsilon - \varepsilon W_{j,k}^\varepsilon E_k^\varepsilon \delta_\theta^2 u_k^\varepsilon - W_{j,k}^\varepsilon E_k^\varepsilon \widehat{H}\left(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon\right) - 2\varepsilon E_k^\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k}. \tag{3.24}$$

Thus the reconstructed density satisfies a semi-discrete analogue of the original density equation, with $R_{j,k}^\varepsilon$ collecting the trait-direction discretization effects. When the numerical Hamiltonian and the conservative flux are evaluated on smooth exact functions, this remainder is consistent with the trait diffusion contribution $\varepsilon^2 \partial_{\theta\theta} n^\varepsilon$.

3.2 Basic structural property

The semi-discrete scheme (3.23) can be shown to be positive-preserving as long as the initial data is nonnegative. The detailed result is summarized as the following lemma.

Lemma 3.1 (Positivity of the amplitude)**.** *Assume that $W^{\varepsilon}_{j,k}(0) \geq 0$ for all j,k. Then the semi-discrete amplitude equation in* (3.21) *preserves nonnegativity, namely*

$$W^{\varepsilon}_{j,k}(t) \geq 0 \qquad \forall\, j,k,\ t \geq 0,$$

as long as the solution exists. Consequently,

$$n^{\varepsilon}_{j,k}(t) \geq 0 \qquad \forall\, j,k,\ t \geq 0.$$

Proof. It is enough to verify the quasi-positivity of the right-hand side of the finite-dimensional ODE system for W^{ε}. From the amplitude equation in (3.21), we can write

$$\varepsilon \frac{\mathrm{d}}{\mathrm{d}t} W^{\varepsilon}_{j,k} = D_k \delta_x^2 W^{\varepsilon}_{j,k} + \varepsilon^2 \delta_\theta^2 W^{\varepsilon}_{j,k} - 2\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}(W^{\varepsilon}, u^{\varepsilon})_{j,k} + W^{\varepsilon}_{j,k} B_{j,k}(t),$$

where

$$B_{j,k}(t) = K_j - \rho^{\varepsilon}_{j,Q}(t) + \mathcal{H}_k(\boldsymbol{\rho}_Q(t)) - 2\varepsilon \delta_\theta^2 u^{\varepsilon}_k(t).$$

Let $W^{\varepsilon} \geq 0$ and suppose that $W^{\varepsilon}_{j,k} = 0$ for some (j,k). Then, for an interior spatial point,

$$\delta_x^2 W^{\varepsilon}_{j,k} = \frac{W^{\varepsilon}_{j+1,k} + W^{\varepsilon}_{j-1,k}}{(\Delta x)^2} \geq 0, \quad \delta_\theta^2 W^{\varepsilon}_{j,k} = \frac{W^{\varepsilon}_{j,k+1} + W^{\varepsilon}_{j,k-1}}{(\Delta \theta)^2} \geq 0.$$

The same conclusion holds at the boundary because of the Neumann ghost values in x-direction and the periodicity in θ-direction. The quasi-positivity assumption on the conservative flux operator implies that

$$-\mathcal{D}_\theta \widehat{\mathcal{F}}(W^{\varepsilon}, u^{\varepsilon})_{j,k} \geq 0.$$

Combining all the inequalities, noticing the fact that $W^{\varepsilon}_{j,k} B_{j,k}(t) = 0$ since we assumed $W^{\varepsilon}_{j,k} = 0$, we finally have that, at this specific (j,k), the following inequality holds,

$$\left. \frac{\mathrm{d}}{\mathrm{d}t} W^{\varepsilon}_{j,k} \right|_{W^{\varepsilon}_{j,k}=0} \geq 0.$$

Thus the vector field points inward on the boundary of the nonnegative cone. The standard invariance criterion for finite-dimensional ODE systems implies the claim. □

3.3 Stationary AP structure under a one-sided stability condition

We now turn to stationary semi-discrete states. The goal is to identify which parts of the continuous rare-mutation structure are retained on a fixed grid. By Lemma 3.1, stationary states obtained from nonnegative amplitudes satisfy

$$W^{\varepsilon}_{j,k} \geq 0, \qquad n^{\varepsilon}_{j,k} = W^{\varepsilon}_{j,k} E^{\varepsilon}_k \geq 0.$$

All estimates below are for such nonnegative reconstructed densities.

Weighted density balance. Define

$$\eta_j^\varepsilon = \Delta\theta \sum_{k=1}^{K} \frac{n_{j,k}^\varepsilon}{D_k}.$$

Multiplying (3.23) by $1/D_k$ and summing over the trait variable gives the stationary weighted balance equation

$$-\delta_x^2 m_j^\varepsilon = \eta_j^\varepsilon \big(K_j - \rho_{j,Q}^\varepsilon\big) + \mathcal{R}_j^{\varepsilon,D}, \tag{3.25}$$

where

$$m_j^\varepsilon = \Delta\theta \sum_{k=1}^{K} n_{j,k}^\varepsilon, \qquad \mathcal{R}_j^{\varepsilon,D} = \Delta\theta \sum_{k=1}^{K} \frac{R_{j,k}^\varepsilon}{D_k}.$$

The term $\mathcal{R}_j^{\varepsilon,D}$ is the discrete analogue of the continuous weighted mutation contribution

$$\varepsilon^2 \int_{\mathbb{T}} \frac{1}{D(\theta)} \partial_{\theta\theta} n^\varepsilon \, d\theta = \varepsilon^2 \int_{\mathbb{T}} n^\varepsilon \partial_{\theta\theta} \left(\frac{1}{D(\theta)}\right) d\theta.$$

At the continuous level this is bounded above by $C\varepsilon^2\rho^\varepsilon$. For the split WKB discretization, however, the discrete operators act on the reconstructed density $n_{j,k}^\varepsilon = W_{j,k}^\varepsilon E_k^\varepsilon$, and an exact chain rule is not available. We isolate the needed property as follows.

Assumption 3.1 (One-sided weighted remainder stability)**.** *There exist constants $C_R > 0$ and $r_h \geq 0$, independent of ε, such that every stationary state under consideration satisfies*

$$\big(\mathcal{R}_j^{\varepsilon,D}\big)_+ \leq C_R m_j^\varepsilon + r_h, \qquad j = 1, \ldots, J. \tag{3.26}$$

This condition only controls the positive part of the weighted defect; a negative contribution is favorable in the maximum-principle estimate. It is a structural stability requirement on the trait discretization, not a consequence of the preceding consistency statement. For a direct density-level conservative discretization, and for the density-compatible WKB variant (**??**), the same type of estimate follows by discrete summation by parts. For the generic split WKB scheme, it should be checked analytically under additional structure or monitored numerically.

We also record the mild stability condition needed for the density reconstruction in the nonlocal reaction term.

Assumption 3.2 (Reconstruction stability)**.** *There exist constants $c_Q, C_Q > 0$ and $e_Q \geq 0$, independent of ε, such that*

$$\rho_{j,Q}^\varepsilon \geq c_Q m_j^\varepsilon - e_Q, \qquad j = 1, \ldots, J, \tag{3.27}$$

$$\rho_{j,Q}^\varepsilon \leq C_Q m_j^\varepsilon + e_Q, \qquad j = 1, \ldots, J. \tag{3.28}$$

For the direct composite quadrature $\rho_{j,Q}^\varepsilon = m_j^\varepsilon$, Assumption 3.2 holds with $c_Q = C_Q = 1$ and $e_Q = 0$.

Proposition 3.1 (Conditional density bound for stationary semi-discrete states)**.** *Let $(u^\varepsilon, W^\varepsilon)$ be a stationary semi-discrete state with nonnegative reconstructed density. Assume that Assumptions 3.1 and 3.2 hold. Then there exist constants $C_m, C_\rho > 0$, independent of ε, such that*

$$0 \leq m_j^\varepsilon \leq C_m, \qquad 0 \leq \rho_{j,Q}^\varepsilon \leq C_\rho, \qquad j = 1, \ldots, J.$$

Proof. Since $n^\varepsilon_{j,k} \geq 0$,

$$\frac{1}{D_{\max}} m^\varepsilon_j \leq \eta^\varepsilon_j \leq \frac{1}{D_{\min}} m^\varepsilon_j.$$

Using (3.25) and (3.26), we obtain

$$-\delta_x^2 m^\varepsilon_j + \eta^\varepsilon_j \rho^\varepsilon_{j,Q} \leq K_{\max} \eta^\varepsilon_j + C_R m^\varepsilon_j + r_h.$$

Let j_* be a point where $M^\varepsilon := \max_j m^\varepsilon_j = m^\varepsilon_{j_*}$. Since $\delta_x^2 m^\varepsilon_{j_*} \leq 0$,

$$\eta^\varepsilon_{j_*} \rho^\varepsilon_{j_*,Q} \leq K_{\max} \eta^\varepsilon_{j_*} + C_R M^\varepsilon + r_h.$$

If $M^\varepsilon \leq e_Q / c_Q$, the bound is immediate. Otherwise, $\rho^\varepsilon_{j_*,Q} \geq c_Q M^\varepsilon - e_Q > 0$, and hence

$$\frac{M^\varepsilon}{D_{\max}} \big(c_Q M^\varepsilon - e_Q \big) \leq \left(\frac{K_{\max}}{D_{\min}} + C_R \right) M^\varepsilon + r_h.$$

This quadratic inequality gives a bound for M^ε independent of ε. The upper estimate for $\rho^\varepsilon_{j,Q}$ then follows from (3.28). □

Stationary phase estimate. The density bound controls the coefficients in the discrete eigenvalue problem and hence the stationary phase equation.

Proposition 3.2 (Stationary phase estimate on a fixed trait grid)**.** *Assume that*

$$0 \leq \rho^\varepsilon_{j,Q} \leq \overline{\rho}, \qquad j = 1, \ldots, J,$$

where $\overline{\rho}$ is independent of ε. Then the discrete effective Hamiltonian defined by (3.20) *satisfies*

$$|\mathcal{H}_k(\boldsymbol{\rho}_Q)| \leq C_{\mathcal{H}}, \qquad k = 1, \ldots, K,$$

where $C_{\mathcal{H}}$ is independent of ε.

Assume further that the stationary phase equation

$$\widehat{H}_G \left(D_\theta^- u^\varepsilon_k, D_\theta^+ u^\varepsilon_k \right) - \varepsilon \delta_\theta^2 u^\varepsilon_k = -\mathcal{H}_k(\boldsymbol{\rho}_Q) \tag{3.29}$$

holds with the Godunov numerical Hamiltonian

$$\widehat{H}_G(p^-, p^+) = -\big((p^-)_- \big)^2 - \big((p^+)_+ \big)^2.$$

Then

$$\Delta\theta \sum_{k=1}^{K} |D_\theta^+ u^\varepsilon_k|^2 \leq C_{\mathcal{H}}.$$

Consequently,

$$\max_k |D_\theta^+ u^\varepsilon_k| \leq \frac{\sqrt{C_{\mathcal{H}}}}{\sqrt{\Delta\theta}}.$$

If the phase is normalized by

$$\max_k u^\varepsilon_k = 0,$$

then

$$\max_k |u^\varepsilon_k| \leq \sqrt{C_{\mathcal{H}}}.$$

All constants are independent of ε, although they may depend on the fixed trait mesh.

Proof. The discrete eigenvalue problem is associated with the symmetric operator

$$-D_k\delta_x^2 - (K_j - \rho_{j,Q}^\varepsilon).$$

Its principal eigenvalue has the Rayleigh representation

$$\mathcal{H}_k(\boldsymbol{\rho}_Q) = \min_{\varphi\neq 0} \frac{D_k\Delta x \sum_j |\delta_x^+\varphi_j|^2 - \Delta x \sum_j (K_j - \rho_{j,Q}^\varepsilon)\varphi_j^2}{\Delta x \sum_j \varphi_j^2}.$$

Since $0 \le K_j \le K_{\max}$ and $0 \le \rho_{j,Q}^\varepsilon \le \overline{\rho}$, we have

$$\mathcal{H}_k(\boldsymbol{\rho}_Q) \ge -K_{\max}.$$

Testing the quotient with a constant vector gives $\mathcal{H}_k(\boldsymbol{\rho}_Q) \le \overline{\rho}$. Thus

$$|\mathcal{H}_k(\boldsymbol{\rho}_Q)| \le \max\{K_{\max}, \overline{\rho}\} =: C_{\mathcal{H}}.$$

Set $q_k^\varepsilon = D_\theta^+ u_k^\varepsilon$. Then $D_\theta^- u_k^\varepsilon = q_{k-1}^\varepsilon$, and

$$-\widehat{H}_G\left(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon\right) = \left((q_{k-1}^\varepsilon)_-\right)^2 + \left((q_k^\varepsilon)_+\right)^2.$$

Summing (3.29) over the periodic trait grid eliminates $\delta_\theta^2 u^\varepsilon$. Therefore

$$\begin{aligned}\Delta\theta \sum_{k=1}^{K} |q_k^\varepsilon|^2 &= -\Delta\theta \sum_{k=1}^{K} \widehat{H}_G\left(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon\right) \\ &= \Delta\theta \sum_{k=1}^{K} \mathcal{H}_k(\boldsymbol{\rho}_Q) \le C_{\mathcal{H}}.\end{aligned}$$

The pointwise slope bound follows from the fixed-grid inequality $|q_k|^2 \le (\Delta\theta)^{-1}\Delta\theta \sum_\ell |q_\ell|^2$. If $\max_k u_k^\varepsilon = 0$, connecting any node to a maximizer along the periodic grid gives

$$|u_k^\varepsilon| \le \Delta\theta \sum_{\ell=1}^{K} |D_\theta^+ u_\ell^\varepsilon| \le \left(\Delta\theta \sum_{\ell=1}^{K} |D_\theta^+ u_\ell^\varepsilon|^2\right)^{1/2}.$$

This proves the claim. □

Formal AP trait-selection limit. We finally pass formally to the fixed-grid rare-mutation limit. The statement is deliberately fixed-grid and conditional: it identifies the limiting selection mechanism once the density and phase estimates above are available.

Proposition 3.3 (Formal AP trait-selection limit)**.** *Assume that the stationary semi-discrete states satisfy Assumptions 3.1 and 3.2, and that the phase is normalized by*

$$\max_k u_k^\varepsilon = 0.$$

Assume also that the stationary phase estimate of Proposition 3.2 applies. Then, up to a subsequence on the fixed grids,

$$\boldsymbol{\rho}_Q^\varepsilon \to \boldsymbol{\rho}^0, \qquad u^\varepsilon \to u^0.$$

Moreover, the limiting phase satisfies the discrete constrained Hamilton–Jacobi system

$$\widehat{H}\left(D_\theta^- u_k^0, D_\theta^+ u_k^0\right) = -\mathcal{H}_k(\boldsymbol{\rho}^0), \qquad \max_k u_k^0 = 0.$$

For the Godunov Hamiltonian $\widehat{H}_G$, every maximizer k_* of u^0 satisfies

$$\mathcal{H}_{k_*}(\boldsymbol{\rho}^0) = 0.$$

If $\mathcal{H}_k(\boldsymbol{\rho}^0)$ has a unique zero at k_m, then $k_* = k_m$. Hence the selected grid trait is identified by

$$\theta_{k_m} \in \arg\max_k u_k^0.$$

Proof. Proposition 3.1 and (3.28) give a uniform bound on $\boldsymbol{\rho}_Q^\varepsilon$. Since the grids are fixed, a subsequence converges to some $\boldsymbol{\rho}^0$. The normalized phase is also uniformly bounded on the fixed trait grid by Proposition 3.2; hence, up to a further subsequence, $u^\varepsilon \to u^0$.

Passing to the limit in the stationary phase equation gives the discrete constrained Hamilton–Jacobi system, because $\varepsilon\delta_\theta^2 u_k^\varepsilon \to 0$ on the fixed grid. For the Godunov Hamiltonian, $\widehat{H}_G \le 0$, and therefore $\mathcal{H}_k(\boldsymbol{\rho}^0) = -\widehat{H}_G(D_\theta^- u_k^0, D_\theta^+ u_k^0) \ge 0$. At a maximizer k_* of u^0,

$$D_\theta^- u_{k_*}^0 \ge 0, \qquad D_\theta^+ u_{k_*}^0 \le 0,$$

so $\widehat{H}_G(D_\theta^- u_{k_*}^0, D_\theta^+ u_{k_*}^0) = 0$. Hence $\mathcal{H}_{k_*}(\boldsymbol{\rho}^0) = 0$. If this zero is unique, the maximizer must be k_m. □

Remark 3.1. *The above result describes the fixed-grid AP structure of the stationary selection state. In the computations, this stationary state is reached through the time-dependent WKB evolution. Thus the evolutionary scheme should be viewed as a numerical relaxation path toward the discrete eigenvalue and constrained Hamilton–Jacobi structure characterized above. This distinction is useful for interpreting the fully discrete method. The semi-discrete analysis identifies the limiting stationary structure, while the fully discrete algorithm evolves u and W toward this structure.*

4 Fully discrete scheme and reconstruction techniques

In this section, we describe the fully discrete WKB method and the reconstruction procedures used to recover the trait concentration profiles. The unknowns advanced in time are the WKB variables u and W. The density, the spatial marginal, and the trait marginal are then recovered from these variables through a reconstruction layer. This layer is part of the numerical method rather than a visualization step. In the small mutation regime, direct evaluation of $W\exp(u/\varepsilon)$ may suffer from overflow, underflow, and exponential amplification of small phase errors. The reconstruction must therefore be stable, mass-consistent, and computationally efficient.

4.1 Fully discrete scheme

Let $t_n = n\tau$. Assume that u^n and W^n are known at time t_n. The reconstructed density and spatial marginal are denoted by

$$n^n = \mathcal{R}_n[W^n, u^n], \qquad \rho^n = \mathcal{R}_\rho[W^n, u^n].$$

After ρ^n is obtained, the discrete effective Hamiltonian $H^n = \mathcal{H}[\rho^n]$ is computed from the corresponding discrete principal eigenvalue problem.

The phase variable is advanced by

$$\frac{\widetilde{u}_k^{n+1} - u_k^n}{\tau} + \widehat{H}(D_\theta^- u_k^n, D_\theta^+ u_k^n) - \varepsilon \delta_\theta^2 \widetilde{u}_k^{n+1} = -H_k^n. \tag{4.30}$$

Here $\widehat{H}$ is the monotone numerical Hamiltonian used in the semi-discrete scheme. The diffusion term in the trait direction is treated implicitly in order to avoid a severe parabolic time-step restriction.

With $\widetilde{u}^{n+1}$ fixed, the amplitude is first advanced by a provisional update. A typical split update takes the form

$$\begin{aligned} &\varepsilon \frac{\widetilde{W}_{j,k}^{n+1} - W_{j,k}^n}{\tau} - D_k \delta_x^2 \widetilde{W}_{j,k}^{n+1} - \varepsilon^2 \delta_\theta^2 \widetilde{W}_{j,k}^{n+1} + 2\varepsilon D_\theta \widehat{F}(\widetilde{W}^{n+1}, \widetilde{u}^{n+1})_{j,k} \\ &\quad = \widetilde{W}_{j,k}^{n+1} \left(K_j - \rho_j^n + H_k^n - 2\varepsilon \delta_\theta^2 \widetilde{u}_k^{n+1} \right). \end{aligned} \tag{4.31}$$

The numerical flux $\widehat{F}$ is chosen consistently with the conservative trait-transport term in the semi-discrete formulation. Density-compatible and Patankar-type variants of the amplitude update may also be used as fully discrete stabilization devices.

After the provisional update, we apply a scalar mass-balance projection. This step is motivated by the density-level identity

$$\varepsilon \frac{dM}{dt} = R(\rho), \qquad M(t) = \int_\Omega \rho(x,t)\, dx, \qquad R(\rho) = \int_\Omega \rho(x,t)(K(x) - \rho(x,t))\, dx.$$

Let $\widetilde{\rho}^{n+1}$ be the spatial marginal reconstructed from $(\widetilde{W}^{n+1}, \widetilde{u}^{n+1})$. We determine a positive scalar λ^{n+1} from a discrete mass-balance relation and set

$$W^{n+1} = \lambda^{n+1} \widetilde{W}^{n+1}, \qquad u^{n+1} = \widetilde{u}^{n+1}.$$

Thus the density-level quantities reconstructed from (W^{n+1}, u^{n+1}) are rescaled by the same factor, while the phase profile is kept fixed. The projection therefore does not move the selected trait location, but it improves the consistency of the reconstructed mass and stabilizes the peak height of the trait marginal. The algebraic form of the scalar projection is given in Appendix B.

Finally, a gauge normalization may be applied to u^{n+1} and W^{n+1}. Since the WKB representation is invariant under the transformation

$$u \mapsto u - c, \qquad W \mapsto W \exp(c/\varepsilon),$$

this normalization improves numerical conditioning without changing the reconstructed density. The stopping criterion is based on gauge-invariant residuals involving the normalized phase and the reconstructed density.

4.2 Reconstruction techniques

The reconstruction layer has three main purposes. First, it reconstructs the solver-level spatial marginal $\rho(x)$, which enters the nonlocal feedback and the effective Hamiltonian. Second, it reconstructs the trait marginal $P(\theta)$, from which the selected trait, peak height, and peak width are extracted. Third, it enables a dual trait-grid implementation that resolves the one-dimensional phase more finely than the space-dependent amplitude, and thereby reduces the dominant computational cost.

Reconstruction of the spatial marginal. For each spatial grid point x_j, define

$$\ell_{j,k}^n = \log\bigl(\max\{W_{j,k}^n, W_{\text{floor}}\}\bigr) + \frac{u_k^n}{\varepsilon}.$$

A stable log-sum-exp reconstruction of the spatial marginal is then given by

$$\widetilde{\rho}_j^n = \exp(m_j^n) \sum_k \omega_k^\theta \exp(\ell_{j,k}^n - m_j^n), \qquad m_j^n = \max_k \ell_{j,k}^n. \tag{4.32}$$

This formula is algebraically equivalent to the direct quadrature of $W \exp(u/\varepsilon)$, but it avoids overflow and underflow caused by large values of u/ε.

When the trait profile is highly concentrated, the quadrature in (4.32) may become sensitive to the trait mesh. In this case, one may use a local Laplace reconstruction near the maximizer of the phase. This reconstruction uses the local curvature of u to approximate the concentrated contribution of the exponential factor. It is useful when the physical width of the concentration layer is smaller than the trait mesh size.

Reconstruction of the trait marginal. The trait marginal is reconstructed from the corrected WKB variables,

$$P_k^n = \int_\Omega W^n(x, \theta_k) \exp(u_k^n/\varepsilon)\, dx.$$

Set

$$A_k^n = \int_\Omega W^n(x, \theta_k)\, dx, \qquad L_k^n = \log P_k^n = \frac{u_k^n}{\varepsilon} + \log\bigl(\max\{A_k^n, A_{\text{floor}}\}\bigr).$$

The reconstruction is performed on $L_k^n = \log P_k^n$, rather than on P_k^n itself. This is important because P_k^n is exponentially sensitive to errors in u_k^n, whereas L_k^n remains a smooth quantity on the WKB scale.

The selected trait is obtained by a two-step local reconstruction. First, L_k^n is reconstructed in the trait direction by a WENO-type interpolation. Second, a local parabolic refinement is applied near the reconstructed maximum. This gives the peak location θ_m^n. Since the scalar mass-balance projection changes only the density scale, it affects the reconstructed peak height but not the phase-based peak location. The same local quadratic approximation also provides stable estimates of the peak height and the peak width. In this way, the narrow peak of $P(\theta)$ can be recovered even when the original trait grid is relatively coarse.

Dual trait-grid implementation. Another important component of the fully discrete method is the dual trait-grid strategy. The phase u and the effective Hamiltonian are computed on a relatively fine trait grid, whereas the space-dependent amplitude W may be evolved on a coarser trait grid. This separation is motivated by the different dimensions and roles of the two WKB variables. The phase u depends only on the trait variable and controls the location and sharpness of the concentration. It is therefore inexpensive to resolve u on a finer trait grid. By contrast, W depends on both space and trait. In d spatial dimensions, the storage and evolution of W scale like

$$O(N_x^d N_\theta^W),$$

up to the cost of the spatial solvers, while the phase variable only carries a one-dimensional trait cost. Hence reducing N_θ^W can lead to a substantial saving in memory and CPU time, especially when the spatial dimension or the spatial resolution is large.

In the dual-grid implementation, the fine-grid phase is used to resolve the trait-scale information and to locate the peak, while the amplitude is stored and evolved on the coarser trait grid needed for the spatial profile. Transfer between the two trait grids is performed within the reconstruction layer. This design preserves the main advantage of the WKB formulation: the concentrated trait profile is resolved through the low-dimensional phase rather than by evolving the full space-trait density on a fine trait mesh.

In summary, the fully discrete method consists of a time-marching step for the WKB variables, a scalar mass-balance projection for density-level consistency, and a reconstruction layer for ρ, $P(\theta)$, and the local peak structure. The log-stabilized reconstruction improves robustness in the small mutation regime, the local peak reconstruction gives accurate selected-trait diagnostics, and the dual-grid strategy reduces the dominant cost associated with the space-dependent amplitude.

5 Numerical experiments

This section describes the numerical experiments used to test the WKB formulation. The tests are organized to match the main assumptions and claims of the method: the structural assumptions used in the stationary AP argument, the emergence of concentration in the trait direction, the accuracy of the reconstructed density when the grid is sufficiently fine, the advantage of the WKB phase on coarse trait grids, standard accuracy checks, and a two-dimensional spatial experiment.

5.1 Common computational setting

Unless otherwise stated, the one-dimensional tests use

$$\Omega = (0,1), \qquad \mathbb{T} = [0,1),$$

with homogeneous Neumann boundary conditions in the spatial variable and periodic boundary conditions in the trait variable. The baseline coefficients used in the current one-dimensional code are

$$K(x) = K_0 - K_1\cos(2\pi x), \qquad K_0 = 1, \qquad K_1 = 0.5, \tag{5.33}$$

$$D(\theta) = D_{\min} + D_1\big(1-\cos(2\pi(\theta-\theta_m))\big), \qquad D_{\min} = 0.2, \qquad D_1 = 0.4, \qquad \theta_m = 0.35. \tag{5.34}$$

Thus the dispersal rate has a unique minimum at θ_m, which is the expected selected trait in the rare-mutation limit. The run scripts specify the initial phase for each test. The two representative choices are a flat phase and a peaked phase,

$$u_k^0 = 0 \qquad \text{or} \qquad u_k^0 = -\alpha_0\big(1-\cos(2\pi(\theta_k-\theta_0))\big), \quad \theta_0 = 0.70, \tag{5.35}$$

with $\alpha_0 = 0.05$ in the non-well-prepared tests and $\alpha_0 = O(\varepsilon)$ in some small-ε refinement tests. The amplitude is usually initialized by

$$W_{j,k}^0 = 1, \qquad n_{j,k}^0 = W_{j,k}^0\exp(u_k^0/\varepsilon). \tag{5.36}$$

When the peaked phase is used, its initial maximizer θ_0 is deliberately placed away from θ_m, so the computation tests selection rather than preservation of a well-prepared peak.

For the WKB computation we use the phase equation in (3.21) together with the selected amplitude update. The density in the nonlocal reaction term is reconstructed from (W^n, u^n). The log-stabilized direct quadrature is

$$\rho_{j,Q}^n = \exp(s_j^n)\Delta\theta\sum_{k=1}^{K}\exp\left(\log\big(\max\{W_{j,k}^n, W_{\text{floor}}\}\big) + \frac{u_k^n}{\varepsilon} - s_j^n\right), \tag{5.37}$$

where

$$s_j^n = \max_k \left\{ \log\bigl(\max\{W_{j,k}^n, W_{\text{floor}}\}\bigr) + \frac{u_k^n}{\varepsilon} \right\}.$$

The code also uses the hybrid Laplace reconstruction described in Subsection **??** and Appendix C; in particular, the small-ε runs can use Laplace reconstruction for ρ, while the trait marginal $P(\theta)$ is reconstructed in logarithmic form for peak localization and width estimation. After each time step, we use the gauge normalization

$$c^n = \max_k u_k^n, \qquad u_k^n \leftarrow u_k^n - c^n, \qquad W_{j,k}^n \leftarrow W_{j,k}^n \exp(c^n/\varepsilon), \tag{5.38}$$

which preserves the reconstructed density $W_{j,k}^n \exp(u_k^n/\varepsilon)$. The fully discrete reconstruction pipeline, residuals, time-step restrictions, mass correction, and dual trait-grid option used in the code are summarized in Subsection **??**.

The raw trait marginal, selected traits, and concentration width are computed as

$$P_k = \int_\Omega n(x, \theta_k)\, dx, \qquad \theta_{\text{dir}} = \theta_{\arg\max_k P_k}, \qquad \theta_{\text{WKB}} = \theta_{\arg\max_k u_k}, \tag{5.39}$$

while the postprocessed small-ε marginal uses the logP-WENO–mass–quadratic reconstruction described in (**??**)–(**??**). In that case the reported peak location is the reconstructed maximizer of $\log P$, with optional parabolic subgrid refinement, rather than the maximizer of a linearly interpolated marginal.

$$d_{\mathbb{T}}(a, b) = \min_{q \in \mathbb{Z}} |a - b + q|, \tag{5.40}$$

$$\sigma_\theta = \left(\frac{\sum_{k=1}^K d_{\mathbb{T}}(\theta_k, \theta_{\text{WKB}})^2 P_k}{\sum_{k=1}^K P_k} \right)^{1/2}. \tag{5.41}$$

For the weighted remainder assumption, we monitor

$$\Gamma_R^\varepsilon = \max_{1 \le j \le J} \frac{(R_j^{\varepsilon,D})_+}{m_j^\varepsilon + 10^{-14}}, \qquad m_j^\varepsilon = \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon, \qquad R_j^{\varepsilon,D} = \Delta\theta \sum_{k=1}^K \frac{R_{j,k}^\varepsilon}{D_k}. \tag{5.42}$$

A bounded value of Γ_R^ε as ε decreases is not a proof of Assumption 3.1, but it is a useful numerical diagnostic for the positive part of the weighted defect. In addition to Γ_R^ε, the fully discrete diagnostics described in Subsection **??** are reported. In particular, the phase residual, amplitude residual, density residual, and the selected trait are monitored separately, since the WKB phase may converge to the selected trait before the full amplitude has reached a numerical steady state. For the direct density solver, the analogous residual is computed with $n^{n+1} - n^n$.

5.2 Test 1: numerical verification of the structural assumptions

The first test checks the structural quantities that appear in the stationary AP argument. We use

$$J = 200, \qquad K = 64, \qquad \varepsilon \in \{5 \times 10^{-2}, 2 \times 10^{-2}, 10^{-2}, 5 \times 10^{-3}\}.$$

Both the split WKB update and the density-compatible update are run. The following quantities are reported:

- $\min_{j,k} W_{j,k}$, to monitor positivity of the amplitude;

- $\max_j \rho_{j,Q}$, to monitor the density bound;
- Γ_R^ε, to test the one-sided weighted remainder condition;
- $\min_k H_k(\rho_Q)$ and the location of this minimum, to check consistency with the selected dispersal minimizer;
- $d_{\mathbb{T}}(\theta_{\mathrm{WKB}}, \theta_m)$, to check trait selection.

The density-compatible update is expected to keep Γ_R^ε controlled by the discrete second difference of $1/D_k$, as explained in Appendix D. For the generic split update, this table is a diagnostic for the conditional assumption.

Table 1: Test 1. Numerical diagnostics for the structural assumptions.

variant	ε	$\min W$	$\max \rho_Q$	Γ_R^ε	θ_{WKB}	$d_{\mathbb{T}}(\theta_{\mathrm{WKB}}, \theta_m)$
split	5×10^{-2}	–	–	–	–	–
split	2×10^{-2}	–	–	–	–	–
split	10^{-2}	–	–	–	–	–
split	5×10^{-3}	–	–	–	–	–
density-compatible	5×10^{-2}	–	–	–	–	–
density-compatible	2×10^{-2}	–	–	–	–	–
density-compatible	10^{-2}	–	–	–	–	–
density-compatible	5×10^{-3}	–	–	–	–	–

5.3 Test 2: long-time concentration in the trait direction

This test checks whether the solution actually concentrates in the trait direction after sufficiently large time. We use

$$J = 200, \qquad K = 128, \qquad \varepsilon \in \{5 \times 10^{-2}, 2 \times 10^{-2}, 10^{-2}\},$$

and store snapshots at several times, for example

$$t \in \{1, 5, 10, 20, 40, 80\}.$$

For each snapshot, we plot the trait marginal $P_k(t)$, the phase $u_k(t)$, and record $\theta_{\mathrm{WKB}}(t)$, $\theta_{\mathrm{dir}}(t)$, and $\sigma_\theta(t)$. The expected behavior is that the maximizer of u moves toward θ_m, while the concentration width σ_θ decreases with time and with ε.

Table 2: Test 2. Long-time concentration diagnostics.

ε	t	$\theta_{\mathrm{WKB}}(t)$	$\theta_{\mathrm{dir}}(t)$	$\sigma_\theta(t)$	$\min_k H_k(\rho_Q(t))$
5×10^{-2}	–	–	–	–	–
2×10^{-2}	–	–	–	–	–
10^{-2}	–	–	–	–	–

5.4 Test 3: small-ε trait-marginal reconstruction and dual-grid WKB advantage

This test focuses on the quantity that becomes most singular when ε is small: the trait marginal $P(\theta)$. The current code uses a very small mutation parameter, for example

$$\varepsilon = 10^{-4}, \qquad J = 40, \qquad T = 2,$$

and compares several WKB configurations with coarse amplitude trait grids. In the dual-grid runs, the amplitude grid has $K_W = 8$ or 16 points, while the phase grid has $K_u = 32$ or 64 points. Thus the phase peak and curvature are resolved on a finer one-dimensional grid, but the space-dependent amplitude is still advanced on a much coarser trait grid.

For each saved snapshot we compare three reconstructions of $P(\theta)$:

raw : compute the nodal marginal and do no peak reconstruction,
basic : interpolate $\log P$ by a standard periodic interpolant,
full : use WENO peak localization in $\log P$, mass normalization, and the phase-amplitude quadratic reconstruction (**??**).

The purpose is to separate two errors that are otherwise mixed together. The first is the error in the WKB variables (W, u); the second is the error made when a very sharp marginal is reconstructed from those variables. Directly interpolating P is not reliable in this regime, because P may be localized between two coarse trait nodes and its height is exponentially sensitive to the phase. Reconstructing $\log P$, locating the peak by WENO, and then imposing the corrected mass stabilizes the peak position and height. The local quadratic fit supplies the peak-width estimate σ_P.

Table 3: Test 3. Small-ε reconstruction of the trait marginal.

K_W	K_u	raw peak	basic peak	full peak	full mass	σ_P	CPU time
8	32	–	–	–	–	–	–
8	64	–	–	–	–	–	–
16	64	–	–	–	–	–	–
64	64	–	–	–	–	–	–

The same postprocessing is also applied to direct density snapshots when they are available, using $\log P_k = \log \int_\Omega n(x, \theta_k)\, dx$. This gives a fair comparison of peak localization while keeping the WKB-specific phase-amplitude curvature information as an additional option for the full WKB reconstruction.

5.5 Test 4: coarse trait grids and AP advantage of the WKB phase

This test demonstrates the main computational advantage of the WKB formulation. We keep ε small and vary the number of trait grid points:

$$J = 200, \qquad \varepsilon = 5 \times 10^{-3}, \qquad K \in \{16, 24, 32, 48, 64, 128\}.$$

A high-resolution direct computation or, when this is too expensive, the limiting trait θ_m, is used as the reference. We compare the selected traits

$$e_{\rm dir} = d_{\mathbb{T}}(\theta_{\rm dir}, \theta_{\rm ref}), \qquad e_{\rm WKB} = d_{\mathbb{T}}(\theta_{\rm WKB}, \theta_{\rm ref}).$$

The expected outcome is that the direct density solver requires a much finer trait grid to localize the narrow peak, whereas the WKB phase remains a smoother object and identifies the concentration location on substantially coarser trait grids. In the dual-grid version we distinguish the cost of the one-dimensional phase grid from the cost of the space-dependent amplitude grid. Increasing K_u improves phase peak localization at a relatively small cost, while increasing K_W multiplies the number of spatial amplitude unknowns. This distinction is minor in one spatial dimension but becomes important in two or more spatial dimensions, where the amplitude solve is the dominant cost.

Table 4: Test 4. Coarse trait-grid comparison and AP advantage.

K	θ_{ref}	θ_{dir}	θ_{WKB}	e_{dir}	e_{WKB}	CPU ratio
16	–	–	–	–	–	–
24	–	–	–	–	–	–
32	–	–	–	–	–	–
48	–	–	–	–	–	–
64	–	–	–	–	–	–
128	–	–	–	–	–	–

5.6 Test 5: accuracy and convergence tests

The fifth test checks standard numerical accuracy. Since closed-form solutions are not available, we compute a reference solution on a refined grid and compare coarser computations with it. For example, use

$$\varepsilon = 2 \times 10^{-2}, \qquad (J_{\mathrm{ref}}, K_{\mathrm{ref}}, \tau_{\mathrm{ref}}) = (400, 256, 2.5 \times 10^{-4}),$$

and compare against coarser triples such as

$$(J, K, \tau) \in \{(100, 64, 10^{-3}), (200, 128, 5 \times 10^{-4}), (300, 192, 3.33 \times 10^{-4})\}.$$

The reported quantities are the errors in ρ, the trait marginal P_k, the phase u_k up to the gauge $\max_k u_k = 0$, and the selected trait. If different trait grids are used, the reference phase and marginal are interpolated periodically in θ. We also perform a time-step sensitivity test for the WKB evolution. This test records E^n_{gi}, its phase and amplitude components, the selected trait θ_{WKB}, and the diagnostics associated with the projective time integrator. The purpose is to separate two questions. The first one is whether the WKB phase correctly captures the fittest trait on a coarse trait grid. The second one is whether the full amplitude and density variables have reached a fully converged steady state. In the very small mutation regime, the first may hold before the second.

Table 5: Test 5. Accuracy against a refined WKB reference solution.

J	K	τ	error in ρ	error in P	trait error
100	64	10^{-3}	–	–	–
200	128	5×10^{-4}	–	–	–
300	192	3.33×10^{-4}	–	–	–

5.7 Test 6: two-dimensional spatial experiment

The last test checks that the WKB strategy is not restricted to one spatial dimension. We solve the model on

$$\Omega = (0,1)^2$$

using a triangular finite-element discretization in $x = (x_1, x_2)$ and the same periodic finite-difference discretization in θ. A representative two-dimensional landscape is

$$K(x_1, x_2) = 1 + 0.35\cos(2\pi x_1)\cos(2\pi x_2) + 0.15\cos(4\pi x_1), \tag{5.43}$$

with the same baseline dispersal function (5.34). Use

$$K_\theta \in \{32, 64\}, \qquad \varepsilon \in \{2\times 10^{-2}, 10^{-2}\}.$$

The outputs are the spatial total density $\rho(x_1, x_2)$, the trait marginal P_k, the phase u_k, and the selected trait θ_{WKB}. This test is mainly qualitative: it should show that the spatial heterogeneity is resolved by the finite-element amplitude/eigenvalue solves, while the concentration location is still captured through the one-dimensional phase variable.

Table 6: Test 6. Two-dimensional spatial WKB experiment.

ε	K_θ	mesh size	θ_{WKB}	$d_{\mathbb{T}}(\theta_{\mathrm{WKB}}, \theta_m)$
2×10^{-2}	32	–	–	–
2×10^{-2}	64	–	–	–
10^{-2}	32	–	–	–
10^{-2}	64	–	–	–

A Numerical Hamiltonians and conservative fluxes

For the monotone theory in Section 3, the numerical Hamiltonian $\widehat{H}(a,b)$ is assumed to be consistent with $-p^2$, nondecreasing in its first argument, and nonincreasing in its second argument. Two standard choices are the Lax–Friedrichs Hamiltonian

$$\widehat{H}_{\mathrm{LF}}(a,b) = -\left(\frac{a+b}{2}\right)^2 + \frac{\alpha}{2}(a-b), \tag{1.44}$$

where α is chosen no smaller than the relevant characteristic speed, and the Godunov Hamiltonian

$$\widehat{H}_G(a,b) = -(a^-)^2 - (b^+)^2, \qquad a^- := \min\{a,0\}, \qquad b^+ := \max\{b,0\}. \tag{1.45}$$

The stationary phase estimate in Proposition 3.2 uses $\widehat{H}_G$.

The conservative flux in the split amplitude equation is associated with

$$F = -W\partial_\theta u.$$

On a uniform periodic trait grid, write

$$D_\theta \widehat{F}(W,u)_{j,k} = \frac{\widehat{F}_{j,k+1/2} - \widehat{F}_{j,k-1/2}}{\Delta\theta}, \qquad a_{k+1/2} = -\frac{u_{k+1} - u_k}{\Delta\theta}.$$

The upwind flux is

$$\widehat{F}^{\mathrm{up}}_{j,k+1/2} = a^+_{k+1/2} W_{j,k} + a^-_{k+1/2} W_{j,k+1}, \qquad a^+ = \max\{a,0\}, \qquad a^- = \min\{a,0\}. \tag{1.46}$$

A Lax–Friedrichs flux is

$$\widehat{F}^{\mathrm{LF}}_{j,k+1/2} = \frac{1}{2} a_{k+1/2}(W_{j,k} + W_{j,k+1}) - \frac{1}{2}\alpha_{k+1/2}(W_{j,k+1} - W_{j,k}), \qquad \alpha_{k+1/2} \geq |a_{k+1/2}|. \tag{1.47}$$

Both fluxes are conservative in the periodic trait variable. The upwind flux also satisfies the quasi-positivity condition used in Lemma 3.1.

Some numerical experiments use high-order one-sided WENO reconstructions for the derivatives entering the Hamiltonian. This is a numerical option for improving resolution of the phase profile. It is not used in the monotonicity argument of Section 3, and the AP estimate is stated for the monotone Hamiltonians above.

B Scalar mass-balance correction

We give the algebraic form of the scalar mass-balance correction used in the fully discrete method. All spatial integrals below are understood in the discrete sense and may be evaluated by any consistent spatial quadrature associated with the fully discrete scheme.

The correction is based on the mass identity

$$\varepsilon \frac{d}{dt} \int_\Omega \rho(x,t)\, dx = \int_\Omega \rho(x,t)\big(K(x) - \rho(x,t)\big)\, dx.$$

Let $\widetilde{W}^{n+1}$, $\widetilde{u}^{n+1}$, and $\widetilde{\rho}^{n+1}$ denote the provisional amplitude, phase, and reconstructed spatial marginal after one time step. The correction rescales only the amplitude,

$$W^{n+1} = \lambda \widetilde{W}^{n+1}, \qquad u^{n+1} = \widetilde{u}^{n+1}, \qquad \lambda > 0.$$

Consequently,

$$\rho^{n+1} = \lambda \widetilde{\rho}^{n+1}, \qquad n^{n+1} = \lambda \widetilde{n}^{n+1}, \qquad P^{n+1} = \lambda \widetilde{P}^{n+1}.$$

Thus the correction changes only the density scale and leaves the phase profile unchanged.

For compactness, define

$$\widetilde{M} = \int_\Omega \widetilde{\rho}^{n+1}\, dx, \qquad \widetilde{S} = \int_\Omega K \widetilde{\rho}^{n+1}\, dx, \qquad \widetilde{Q} = \int_\Omega (\widetilde{\rho}^{n+1})^2\, dx,$$

and

$$M^n = \int_\Omega \rho^n\, dx, \qquad R^n = \int_\Omega \rho^n (K - \rho^n)\, dx.$$

If the trapezoidal discretization is applied in time, λ is determined by

$$\varepsilon \frac{\lambda \widetilde{M} - M^n}{\tau} = \frac{1}{2}\left[R^n + \lambda \widetilde{S} - \lambda^2 \widetilde{Q}\right].$$

Equivalently,

$$\mathcal{A}_{\mathrm{TR}} \lambda^2 + \mathcal{B}_{\mathrm{TR}} \lambda + \mathcal{C}_{\mathrm{TR}} = 0,$$

where

$$\mathcal{A}_{\mathrm{TR}} = \frac{\tau}{2}\widetilde{Q}, \qquad \mathcal{B}_{\mathrm{TR}} = \varepsilon \widetilde{M} - \frac{\tau}{2}\widetilde{S}, \qquad \mathcal{C}_{\mathrm{TR}} = -\varepsilon M^n - \frac{\tau}{2} R^n.$$

A backward Euler version is obtained from

$$\varepsilon\frac{\lambda\widetilde{M}-M^n}{\tau}=\lambda\widetilde{S}-\lambda^2\widetilde{Q},$$

or equivalently

$$\mathcal{A}_{\mathrm{BE}}\lambda^2+\mathcal{B}_{\mathrm{BE}}\lambda+\mathcal{C}_{\mathrm{BE}}=0,$$

with

$$\mathcal{A}_{\mathrm{BE}}=\tau\widetilde{Q},\qquad \mathcal{B}_{\mathrm{BE}}=\varepsilon\widetilde{M}-\tau\widetilde{S},\qquad \mathcal{C}_{\mathrm{BE}}=-\varepsilon M^n.$$

The positive root is selected in both cases. When the provisional reconstruction already satisfies the discrete mass balance, the correction factor is close to one. In the small-mutation regime, this scalar projection helps stabilize the total mass and the peak height of the trait marginal without changing the phase-based peak location.

C Density and trait-marginal reconstruction

We summarize the reconstruction procedures used to recover density-level quantities from the WKB variables. The basic stable reconstruction of ρ is the log-stabilized quadrature. Define

$$\ell_{j,k}=\log\bigl(\max\{W_{j,k},W_{\mathrm{floor}}\}\bigr)+\frac{u_k}{\varepsilon}.$$

Then

$$\rho_j=\sum_k\omega_k\exp(\ell_{j,k})$$

is evaluated by a row-wise log-sum-exp formula. This is algebraically equivalent to the composite quadrature

$$\rho_{j,Q}=\Delta\theta\sum_k W_{j,k}\exp(u_k/\varepsilon),$$

but avoids underflow and overflow when ε is small.

A local Laplace reconstruction may be used near a nondegenerate maximizer of the phase. Let k_* be a nodal maximizer of u_k, and set

$$d_{\theta\theta}u_{k_*}=\frac{u_{k_*+1}-2u_{k_*}+u_{k_*-1}}{(\Delta\theta)^2}.$$

When $d_{\theta\theta}u_{k_*}<0$, a three-point parabolic refinement gives

$$z_*=-\frac{\Delta\theta}{2}\frac{u_{k_*+1}-u_{k_*-1}}{u_{k_*+1}-2u_{k_*}+u_{k_*-1}},\qquad \theta_*=(\theta_{k_*}+z_*)\bmod 1.$$

Interpolating W and u locally at θ_*, one obtains the Laplace approximation

$$\rho_j^{\mathrm{Lap}}=W(x_j,\theta_*)\exp\left(\frac{u(\theta_*)}{\varepsilon}\right)\left(\frac{2\pi\varepsilon}{|d_{\theta\theta}u_{k_*}|}\right)^{1/2}. \tag{3.48}$$

If the curvature is not negative enough or the local amplitude is not positive, the method falls back to the log-stabilized quadrature.

The trait marginal is reconstructed in logarithmic form. First define

$$A_k=\int_\Omega W(x,\theta_k)\,dx,$$

and then

$$L_k = \log P_k = \frac{u_k}{\varepsilon} + \log\bigl(\max\{A_k, A_{\text{floor}}\}\bigr). \tag{3.49}$$

Interpolation and peak localization are applied to $L = \log P$, not to P. A WENO interpolation of L_k is used on a fine periodic search grid, followed by a local parabolic refinement to obtain the peak location.

After the peak location has been found, a local quadratic reconstruction gives the peak width. If

$$u(\theta_* + z) \simeq \alpha_0 + \alpha_1 z + \alpha_2 z^2, \qquad \log A(\theta_* + z) \simeq \beta_0 + \beta_1 z + \beta_2 z^2,$$

then

$$\log P(\theta_* + z) \simeq \left(\frac{\alpha_0}{\varepsilon} + \beta_0\right) + \left(\frac{\alpha_1}{\varepsilon} + \beta_1\right) z + \left(\frac{\alpha_2}{\varepsilon} + \beta_2\right) z^2.$$

When $\alpha_2/\varepsilon + \beta_2 < 0$, the fitted peak width is

$$\sigma_P \simeq \frac{1}{\sqrt{-2(\alpha_2/\varepsilon + \beta_2)}}. \tag{3.50}$$

The reconstructed shape may then be normalized by a prescribed mass M_{target},

$$P_{\text{rec}}(\theta) = M_{\text{target}} \frac{\exp(\widehat{L}(\theta) - \max_\eta \widehat{L}(\eta))}{\int_{\mathbb{T}} \exp(\widehat{L}(\eta) - \max_\xi \widehat{L}(\xi))\, d\eta}. \tag{3.51}$$

This normalization stabilizes the reconstructed peak height after the location and width have been determined.

For the dual trait-grid implementation, let Θ_u be the fine phase grid and Θ_W the coarse amplitude grid. The amplitude is first interpolated to the phase grid,

$$W_U = I_{W\to u} W_c,$$

and the log-stabilized quadrature is then performed on Θ_u:

$$\rho_j = \sum_{\theta_k \in \Theta_u} \omega_k \exp\left(\log\bigl(\max\{(W_U)_{j,k}, W_{\text{floor}}\}\bigr) + \frac{u_k}{\varepsilon}\right).$$

Thus the phase and peak localization are resolved on the fine one-dimensional trait grid, while the space-dependent amplitude is stored and evolved on the coarser trait grid.

D A sufficient condition for the one-sided remainder bound

We give a simple situation in which Assumption **??** can be verified. Suppose that the reconstructed density satisfies the conservative semi-discrete density equation

$$\varepsilon \frac{\mathrm{d}}{\mathrm{d}t} n^\varepsilon_{j,k} - D_k \delta_x^2 n^\varepsilon_{j,k} - \varepsilon^2 \delta_\theta^2 n^\varepsilon_{j,k} = \bigl(K_j - \rho^\varepsilon_{j,Q}\bigr) n^\varepsilon_{j,k}. \tag{4.52}$$

Then, in the notation of (3.23),

$$R^\varepsilon_{j,k} = \varepsilon^2 \delta_\theta^2 n^\varepsilon_{j,k}.$$

Therefore

$$R^{\varepsilon,D}_j = \varepsilon^2 \Delta\theta \sum_k \frac{\delta_\theta^2 n^\varepsilon_{j,k}}{D_k}.$$

Periodic summation by parts gives

$$R_j^{\varepsilon,D} = \varepsilon^2 \Delta\theta \sum_k n_{j,k}^{\varepsilon} \delta_\theta^2 \left(\frac{1}{D_k}\right).$$

Hence, for $0 < \varepsilon \leq 1$,

$$\left(R_j^{\varepsilon,D}\right)_+ \leq C_{D,h} m_j^{\varepsilon}, \qquad C_{D,h} = \max_k \left|\delta_\theta^2 \left(\frac{1}{D_k}\right)\right|, \qquad m_j^{\varepsilon} = \Delta\theta \sum_k n_{j,k}^{\varepsilon}.$$

This verifies Assumption **??** with $C_R = C_{D,h}$ and $r_h = 0$. If D is smooth and the standard centered periodic difference is used, $C_{D,h}$ remains bounded under regular trait-grid refinement.

E Direct density solver and two-dimensional finite-element realization

For comparison, the original density equation can be discretized by the linear implicit update

$$\varepsilon \frac{n_{j,k}^{n+1} - n_{j,k}^n}{\tau} - D_k \delta_x^2 n_{j,k}^{n+1} - \varepsilon^2 \delta_\theta^2 n_{j,k}^{n+1} = n_{j,k}^{n+1}(K_j - \rho_j^n), \qquad \rho_j^n = \Delta\theta \sum_k n_{j,k}^n. \tag{5.53}$$

A positivity-oriented Patankar variant is

$$\varepsilon \frac{n_{j,k}^{n+1} - n_{j,k}^n}{\tau} - D_k \delta_x^2 n_{j,k}^{n+1} - \varepsilon^2 \delta_\theta^2 n_{j,k}^{n+1} = (K_j - \rho_j^n)_+ n_{j,k}^n - (K_j - \rho_j^n)_- n_{j,k}^{n+1}. \tag{5.54}$$

The same spatial and trait grids are used for the direct and WKB solvers in the one-dimensional comparison tests.

For the two-dimensional spatial experiment, the spatial Laplacian and the eigenvalue problem are discretized by linear finite elements on a triangular mesh. For each trait node θ_k, the discrete effective Hamiltonian is computed from

$$A_k(\rho_Q) N^{(k)} = H_k(\rho_Q) M N^{(k)}, \tag{5.55}$$

where M is the finite-element mass matrix and

$$(A_k)_{ab} = D_k \int_\Omega \nabla\varphi_a \cdot \nabla\varphi_b \, dx - \int_\Omega (K(x) - \rho_Q(x)) \varphi_a \varphi_b \, dx. \tag{5.56}$$

The Neumann boundary condition is natural in this formulation. The phase equation remains one-dimensional in θ, while the amplitude equation is solved for all spatial finite-element degrees of freedom and trait nodes.

References

[1] Luis Almeida, Benoît Perthame, and Xinran Ruan. An asymptotic preserving scheme for capturing concentrations in age-structured models arising in adaptive dynamics. *Journal of Computational Physics*, 464:111335, 2022.

[2] Anton Arnold, Naoufel Ben Abdallah, and Claudia Negulescu. WKB-based schemes for the oscillatory 1d Schrödinger equation in the semiclassical limit. *SIAM Journal on Numerical Analysis*, 49(4):1436–1460, 2011.

[3] Guy Barles and Benoît Perthame. Concentrations and constrained Hamilton–Jacobi equations arising in adaptive dynamics. In *Recent Developments in Nonlinear Partial Differential Equations*, volume 439 of *Contemporary Mathematics*, pages 57–68. American Mathematical Society, 2007.

[4] Emeric Bouin and Sepideh Mirrahimi. A Hamilton–Jacobi limit for a model of population structured by space and trait. *Communications in Mathematical Sciences*, 13(6):1431–1452, 2015.

[5] Laurent Desvillettes, Pierre-Emmanuel Jabin, Stéphane Mischler, and Gaël Raoul. On selection dynamics for continuous structured populations. *Communications in Mathematical Sciences*, 6(3):729–747, 2008.

[6] Odo Diekmann, Pierre-Emmanuel Jabin, Stéphane Mischler, and Benoît Perthame. The dynamics of adaptation: an illuminating example and a Hamilton–Jacobi approach. *Theoretical Population Biology*, 67(4):257–271, 2005.

[7] Jack Dockery, Vivian Hutson, Konstantin Mischaikow, and Mark Pernarowski. The evolution of slow dispersal rates: a reaction diffusion model. *Journal of Mathematical Biology*, 37:61–83, 1998.

[8] Wenrui Hao, King-Yeung Lam, and Yuan Lou. Concentration phenomena in an integro-PDE model for evolution of conditional dispersal. *Indiana University Mathematics Journal*, 68(3):881–923, 2019.

[9] Alan Hastings. Can spatial variation alone lead to selection for dispersal? *Theoretical Population Biology*, 24(3):244–251, 1983.

[10] Hélène Hivert. A first-order asymptotic preserving scheme for front propagation in a one-dimensional kinetic reaction–transport equation. *Journal of Computational Physics*, 367:118–146, 2018.

[11] Shi Jin. Asymptotic-preserving schemes for multiscale physical problems. *Acta Numerica*, 31:415–489, 2022.

[12] Jannis Körner, Anton Arnold, and Kirian Döpfner. WKB-based scheme with adaptive step size control for the Schrödinger equation in the highly oscillatory regime. *Journal of Computational and Applied Mathematics*, 404:113905, 2022.

[13] King-Yeung Lam. Stability of Dirac concentrations in an integro-PDE model for evolution of dispersal. *Calculus of Variations and Partial Differential Equations*, 56:79, 2017.

[14] King-Yeung Lam, Yuan Lou, and Benoît Perthame. A Hamilton–Jacobi approach to evolution of dispersal. *Communications in Partial Differential Equations*, 48(1):86–118, 2023.

[15] Alexander Lorz, Sepideh Mirrahimi, and Benoît Perthame. Dirac mass dynamics in multidimensional nonlocal parabolic equations. *Communications in Partial Differential Equations*, 36(6):1071–1098, 2011.

[16] Songting Luo and Nicholas Payne. An asymptotic method based on a Hopf–Cole transformation for a kinetic BGK equation in the hyperbolic limit. *Journal of Computational Physics*, 341:295–312, 2017.

[17] Sepideh Mirrahimi and Benoît Perthame. Asymptotic analysis of a selection model with space. *Journal de Mathématiques Pures et Appliquées*, 104(6):1108–1118, 2015.

[18] Benoît Perthame and Panagiotis E. Souganidis. Rare mutations limit of a steady state dispersal evolution model. *Mathematical Modelling of Natural Phenomena*, 11(4):154–166, 2016.

[19] Ophélie Ronce. How does it feel to be like a rolling stone? ten questions about dispersal evolution. *Annual Review of Ecology, Evolution, and Systematics*, 38:231–253, 2007.